



**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## **Experiment 20. Information Retrieval using TF-IDF**

### **Program**

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
documents = [
```

```
    "Artificial Intelligence is the future.",
```

```
    "Machine Learning is a part of AI.",
```

```
    "Python is used for Artificial Intelligence."
```

```
]
```

```
query = input("Enter search query: ")
```

```
vectorizer = TfidfVectorizer()
```

```
tfidf = vectorizer.fit_transform(documents + [query])
```

```
scores = (tfidf[:-1] * tfidf[-1].T).toarray()
```

```
ranking = sorted(
```

```
    enumerate(scores),
```

```
    key=lambda x: x[1],
```

```
    reverse=True
```

```
)
```

```
print("\nDocument Ranking")
```

```
for index, score in ranking:
```

```
    print("Document", index + 1, "Score =", score[0])
```

## Output

```
ranking = sorted(  
    enumerate(scores),  
    key=lambda x: x[1],  
    reverse=True  
)  
  
print("\nDocument Ranking")  
  
for index, score in ranking:  
    print("Document", index + 1, "Score =", score[0])
```

... Enter search query: Artificial Intelligence

Document Ranking  
Document 1 Score = 0.5028664815427534  
Document 3 Score = 0.43929881371296026  
Document 2 Score = 0.0

